

1. Project Overview

- **Purpose:** A ReactJS application for transport organizations to track consignments, manage logistics, and administer users.
- **Scope:** Provides authentication, dashboard with search/filter, consignment CRUD operations, admin panel for user management, and global toast notifications.
- **Key Features:**
 - Authentication with role-based access (admin vs user).
 - Dashboard with search and status filters.
 - Consignment detail view with logistics fields.
 - Consignment form with validation and CRUD.
 - Admin panel with user CRUD and role updates.
 - Global toast notifications for success/error/info.
 - Pink-themed UI for consistent branding.

2. Tech Stack

- **Frontend:** ReactJS with Vite
- **Routing:** React Router
- **State Management:** Context API (`AuthContext`, `ToastContext`)
- **Styling:** Plain CSS with pink palette
- **API Layer:** Axios/fetch with mock `Data.jsx` acting as database
- **Package Manager:** Yarn

3. Folder Structure

Code

src/

```
|— components/      # Reusable UI components (Layout, ProtectedRoute, AdminRoute)
|— pages/           # Route-based views (Login, Dashboard, ConsignmentDetail,
ConsignmentForm, AdminPanel, NotFound)
|— services/        # API layer and Data.jsx mock database
|— styles/          # CSS files and theme variables
|— context/         # AuthContext and ToastContext
|— App.jsx
|— main.jsx
```

4. Setup Instructions

Clone the repository:

bash

```
git clone https://github.com/your-org/consignment-tracker.git
```

```
cd consignment-tracker
```

1.

Install dependencies:

bash

```
yarn install
```

2.

Run development server:

bash

```
yarn dev
```

3.

4. Environment variables:

- `.env.development` →

```
VITE_API_BASE_URL=http://localhost:4000/api
```

- `.env.production` →

```
VITE_API_BASE_URL=https://api.consignmenttracker.com
```

5. Authentication

- Token simulation with `localStorage`.
- `AuthContext` manages login/logout and persists user state.
- Role-based access:
 - `ProtectedRoute` for authenticated users.
 - `AdminRoute` for admin-only pages.

6. Consignment Management

- **Dashboard:** Lists consignments with search and status filter.
- **Detail View:** Shows logistics fields (origin, destination, reference, status).
- **Form:** Validates inputs and creates consignments.
- **CRUD:** Implemented in `api.js` using `Data.jsx` mock database.

7. Admin Panel

- **User Management:**

- List users.
 - Create new users.
 - Update roles (user/admin).
 - Delete users.
- **System Logs:** Actions recorded in `Data.jsx`.

8. Toast Notifications

- Global `ToastContext` provides `addToast(message, type)`.
- Types: `success`, `error`, `info`.
- Styled with pink palette, auto-dismiss after 3 seconds.

9. Styling

- **Theme Variables:** Defined in `theme.css`.
- **Pink Palette:**
 - Primary: `#e91e63`
 - Dark: `#c2185b`
 - Accent: `#f8bbd0`
 - Background: `#fff0f6`
- Consistent use across Layout, Dashboard, Forms, AdminPanel.

10. Future Roadmap

- Backend integration with real API.
- CI/CD pipeline setup.
- Unit tests with Vitest/Jest.
- Deployment on Netlify/Vercel.
- Internationalization (i18n).
- Mobile-first enhancements.